

Database Systems	
Lab #:	07
Lab Title:	Introduction to SQL. Designing database, table and insertion of data through SQL.
Submitted by:	
Names	Registration #
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Rubrics name & number					Marks	
					In-Lab	Post-Lab
Engineering Knowledge	R2: Use of Engineering Knowledge and follow Experiment Procedures: Ability to follow experimental procedures, control variables, and record procedural steps on lab report.					
Problem Analysis	R6: Experimental Data Analysis: Ability to interpret findings, compare them to values in the literature, identify weaknesses and limitations.					
Design	R8: Best Coding Standards: Ability to follow the coding standards and programming practices.					
Modern Tools Usage	R9: Understand Tools: Ability to describe and explain the principles behind and applicability of engineering tools.					
Individual and Teamwork	R12: Individual Work Contributions: Ability to carry out individual responsibilities.					
	R13: Management of Team Work: Ability to appreciate, understand and work with multidisciplinary team members.					
Rubrics #	R2	R6	R8	R9	R12	R13
In Lab						
Post- Lab						

Lab No:07

Title: “Installation of MySQL Workbench, Introduction to SQL (structured Query Language). Designing database, table and insertion of data through SQL”.

Introduction:

SQL is a transform-oriented language with 2 major components:

- A DDL for defining the database structure.
- A DML for retrieving and updating data.

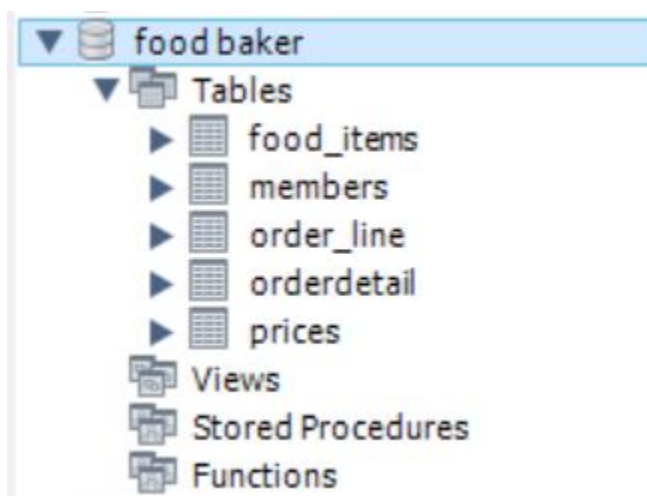
Objective:

- Database language should allow user to:
 - o create the database and relation structures.
 - o perform insertion, modification, deletion of data from relations.
 - o perform simple and complex queries.
- Must perform these tasks with minimal user effort and command structure/syntax must be easy to learn.
- Portable.

In lab and Post lab tasks:

Task 1:

Create the database name FoodBakers.



Task 2:

Using SQL, create the tables identified below in the following order:

DATABASE TABLES:

- Campus (CampusID, CampusName, Street, City, State, Zip, Phone, CampusDiscount).

Table:



Table Name: Schema: **university**

Charset/Collation: Engine:

Comments:

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expre
 CampusID	INT	☒	☒	☐	☐	☐	☐	☐	☐	
 CampusName	VARCHAR(300)	☐	☒	☐	☐	☐	☐	☐	☐	
 Street	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
 City	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
 State	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
 Zip	INT	☐	☒	☐	☐	☐	☐	☐	☐	
 Phone	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
 CampusDiscount	DECIMAL(10,2)	☐	☒	☐	☐	☐	☐	☐	☐	
 Year_of_joining	INT	☐	☒	☐	☐	☐	☐	☐	☐	
 Email	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
		☐	☐	☐	☐	☐	☐	☐	☐	

Insertion of data:

Result Grid  Filter Rows: Edit:    Export/Import:   Wrap Cell Content: 

	CampusID	CampusName	Street	City	State	Zip	Phone	CampusDiscount	Year_of_joining	Email
▶	1	Sumaira Safeer	Street 3	Bhakkar	Punjab	44022	0305-3578543	44.66	2022	SumairaSafeer@gmail.com
	2	Hannah Zoe	Main Street 4	Attock	KPK	44000	0333-6853234	45.99	2024	HannahZoe@gmail.com
	3	Rohan Malik	Street 4	Lahore	Punjab	20544	0300-3468643	5534.87	2022	RohanMalik@gmail.com
	4	Saamia Zee	Main Street 2	Mianwali	Punjab	22276	0343-7531435	987.43	2023	SaamiaZee@gmail.com
	5	Subhana Hashim	Street 6	Sargodah	Punjab	22334	0344-7532685	76.64	2020	SubhanaHashim@gmail.com
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

campus 1 x 

- Position (PositionID, Position, YearlyMembershipFee).

Table:



Table Name:
 Schema: **university**

Charset/Collation:
 Engine:

Comments:

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G
 PositionID	INT	☒	☒	☐	☐	☐	☐	☐	☐
 Position	VARCHAR(15)	☐	☒	☐	☐	☐	☐	☐	☐
 YearlyMembership	FLOAT	☐	☒	☐	☐	☐	☐	☐	☐

Insertion of data:

Result Grid

Filter Rows:

Edit:   

Export/Import:  

	PositionID	Position	YearlyMembership
▶	100	Student	1
	200	Student	4
	300	Student	3
	400	Student	5
	500	Student	2
•	NULL	NULL	NULL

position 1 x

Output

Action Output

#	Time	Action
1	20:53:58	SELECT * FROM university.position LIMIT 0, 1000

- Members (MemberID, LastName, FirstName, CampusAddress, CampusPhone, CampusID, PositionID, ContractDuration).
FK CampusID --> Campus(CampusID) PositionID --> Position(PositionID).

Table:

Table Name: member		Schema: university								
Charset/Collation: utf8mb4 utf8mb4_0900_ai_c		Engine: InnoDB								
Comments:										
Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	D
MemberID	INT	☒	☒	☐	☐	☐	☐	☐	☐	
FirstName	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
LastName	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
Address	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
CampusPhone	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
CampusID	INT	☐	☒	☐	☐	☐	☐	☐	☐	
PositionID	INT	☐	☒	☐	☐	☐	☐	☐	☐	

Insertion of data:

MemberID	FirstName	LastName	Address	CampusPhone	CampusID	PositionID	ContractDuration
11	Hannah	Zoe	Attock	0303-46829279	3	100	5
22	Sumaira	Safeer	Bhakkar	0333-62891229	5	200	3
33	Rohan	Malik	Multan	0344-56457325	1	300	7
44	Subhana	Hashim	Faislabad	0300-68584356	4	400	2
55	Saamia	Zee	Lahore	0333-36455785	2	500	4
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

member 1 x

Output

Action Output

#	Time	Action	Message
1	20:58:21	SELECT * FROM university.member LIMIT 0, 1000	5 row(s) r

- Prices (FoodItemTypeID, MealType, MealPrice).

Table:

	Table Name:	prices	Schema:	food baker
	Charset/Collation:	utf8mb4 utf8mb4_0900_ai_ci	Engine:	InnoDB
	Comments:			

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Exp
 FoodItemsType_ID	INT	☒	☒	☐	☐	☐	☐	☐	☐	
 Meal_Type	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
 Meal_Price	DECIMAL(10,2)	☐	☒	☐	☐	☐	☐	☐	☐	

Insertion of data:

FoodItemsType_ID	Meal_Type	Meal_Price
1	Lunch	7.98
2	Dinner	45.88
3	Dinner	12.98
4	Breakfast	23.87
5	Lunch	100.70
NULL	NULL	NULL

prices 1 ×

Output

Action Output

#	Time	Action
1	20:58:55	SELECT * FROM 'food baker'.prices LIMIT 0, 1000

- **FoodItems (FoodItemID, FoodItemName, FoodItemTypeID).**
FK FoodItemTypeID --> Prices(FoodItemTypeID).

Table:



Table Name:

food_items

Schema:

food baker

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
 FoodItems_ID	INT	☒	☒	☐	☐	☐	☐	☐	☐	
 FoodItems_Name	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
 FoodItemsType_ID	INT	☐	☒	☐	☐	☐	☐	☐	☐	
		☐	☐	☐	☐	☐	☐	☐	☐	

Insertion of data:

Result Grid




Filter Rows:

Edit:




	FoodItems_ID	FoodItems_Name	FoodItemsType_ID
▶	1	Burger	10
	2	Pizza	20
	3	Roll	30
	4	Salid	40
	5	Biryani	50
•	NULL	NULL	NULL

food_items 1 ×

Output

Action Output

#	Time	Action
1	20:45:23	SELECT * FROM 'food baker'.food_items LIMIT 0, 1000

- OrdersDetails (OrderID, MemberID, OrderDate) .

FK MemberID --> Members(MemberID).

Table:

	Table Name:	orderdetail	Schema:	food baker
	Charset/Collation:	utf8mb4 utf8mb4_0900_ai_ci	Engine:	InnoDB
	Comments:			

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
 Orders_ID	INT	☒	☒	☐	☐	☐	☐	☐	☐	
 Member_ID	INT	☐	☒	☐	☐	☐	☐	☐	☐	
 Order_Date	DATE	☐	☒	☐	☐	☐	☐	☐	☐	

Insertion of data:

Result Grid

Filter Rows:

Edit:   

Export/Import:

	Orders_ID	Member_ID	Order_Date
▶	11	13	2023-09-05
	22	11	2023-06-08
	33	12	2024-12-23
	44	14	2023-10-04
	55	15	2024-03-15
✱	NULL	NULL	NULL

orderdetail 1 x

Output

Action Output

#

Time

Action

✓

1

20:59:21

SELECT * FROM 'food baker'.orderdetail LIMIT 0, 1000

- OrderLine (OrderID, FoodItemsID, Quantity).

FK OrderID --> Orders(OrderID) FoodItemsID --> FoodItems(FoodItemID).

Table:

 Table Name: order_line Schema: food baker										
Charset/Collation: utf8mb4 utf8mb4_0900_ai_ci	Engine: InnoDB									
Comments:										
Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
 Order_ID	INT	☒	☒	☐	☐	☐	☐	☐	☐	
 FoodItems_ID	INT	☒	☒	☐	☐	☐	☐	☐	☐	
 Quantity	INT	☐	☒	☐	☐	☐	☐	☐	☐	
 Brand	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	

Insertion of data:

Result Grid

Filter Rows:

Edit: 

Export/Import: 

	Order_ID	FoodItems_ID	Quantity	Brand
▶	11	1	5	Gucci
	22	2	2	Zulbari
	33	4	6	J.
	44	3	3	Amfti
	55	5	4	GulAhmed
✱	NULL	NULL	NULL	NULL

order_line 1 x

Output

Action Output

#	Time	Action
✓ 1	20:48:29	SELECT * FROM 'food baker'.order_line LIMIT 0, 1000